Himaayah Schools
ACADEMY

FREE GUIDE

وَجَعَلْنَا مِنَ الْمَاءِ كُلَّ شَيْءٍ حَيٍّ

*"And We made from water every living thing." (al-Anbiyāʾ 21:30)*

# JAMB Biology

A UTME study guide: the syllabus, key facts, genetics made simple, worked examples and a practice test.

بِسْمِ اللَّهِ الرَّحْمٰنِ الرَّحِيمِ

# Before you begin

UTME Biology is a wide subject, but most questions test clear facts and understanding: what a structure does, how a process works, and how organisms are related. Genetics and ecology bring a few short calculations. This guide gives you the syllabus in five areas, the key facts to know, genetics step by step, nine worked examples, common traps, an 8-week plan and a practice test.

| The UTME at a glance | |
|---|---|
| Subjects | Use of English and three subjects for your course. Biology is required for many science, medical and engineering courses. |
| Questions | Use of English 60; each other subject 40; 180 questions in all, on computer (CBT). |
| Time | About two hours for all four papers, so roughly 40 seconds a question. You move freely between subjects. |
| Marking | Each subject is scored out of 100, for a total out of 400. There is no negative marking: never leave a question blank. |

### Check the official details

Exam rules and syllabuses can change from year to year. Always check the current JAMB brochure and the official Biology syllabus on the JAMB website or e-facility portal, and follow them where they differ from this guide.

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## Part 1 The syllabus at a glance

Study from the **JAMB syllabus**, not only from your school notes: every UTME question comes from a topic on it. The topics fall into these areas:

| Area | Topics |
|---|---|
| Variety of organisms | Characteristics of living things; the cell and its organelles; levels of organisation; classification: kingdoms Monera, Protista, Fungi, Plantae and Animalia, with examples; adaptations; viruses |
| Form and functions | Nutrition in plants (photosynthesis, mineral requirements) and animals (food classes, digestion, enzymes); transport; respiration; excretion; support and movement; reproduction; growth; coordination: nervous system, sense organs, hormones in animals and plants |
| Ecology | Ecological factors and habitats; food chains, webs and pyramids; energy flow; nutrient cycles (carbon, nitrogen, water); population studies; symbiosis; conservation; pollution; pests and diseases |
| Heredity and variation | Variation; Mendel's laws; monohybrid crosses; blood groups; sex determination and sex-linked traits; the sickle-cell genotype; applications of genetics in agriculture and medicine |
| Evolution | Theories of evolution (Lamarck, Darwin); evidence for evolution; adaptation and natural selection |

Form and functions is the largest area. Do not neglect ecology and genetics, which appear every year and are among the easiest marks once understood.

## Part 2 Key facts at a glance

### The cell

| Organelle | Function | Found in |
|---|---|---|
| Nucleus | Controls the cell; contains the chromosomes (DNA) | Plant and animal cells |
| Mitochondrion | Aerobic respiration: releases energy (ATP) | Plant and animal cells |
| Chloroplast | Photosynthesis | Plant cells only |
| Cell wall | Cellulose; gives shape and support | Plant cells (not animal cells) |
| Cell membrane | Controls what enters and leaves | All cells |
| Ribosome | Protein synthesis | All cells |
| Vacuole | Large and permanent in plants; stores cell sap | Mainly plant cells |

### Processes and equations

| Process | Summary | Note |
|---|---|---|
| Photosynthesis | $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (light, chlorophyll) | Oxygen comes from water |
| Aerobic respiration | $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy | In mitochondria |
| Anaerobic: yeast | Glucose → ethanol + $CO_2$ + a little energy | Fermentation |
| Anaerobic: muscle | Glucose → lactic acid + a little energy | Causes muscle fatigue |

## Digestion and food tests

| Item | Detail | Result or product |
|---|---|---|
| Salivary amylase (ptyalin) | Mouth: starch | Maltose |
| Pepsin | Stomach (acid): protein | Polypeptides |
| Trypsin, lipase, amylase | Small intestine (pancreatic juice) | Peptides; fatty acids and glycerol; maltose |
| Bile | From the liver, stored in the gall bladder: emulsifies fats | No enzyme |
| Starch test | Iodine solution | Blue-black |
| Reducing sugar test | Benedict's or Fehling's solution, heated | Brick-red |
| Protein test | Biuret test (or Millon's reagent) | Violet (or brick-red with Millon's) |
| Fat test | Sudan III, or grease spot on paper | Red layer; translucent spot |

## Hormones

| Hormone | Gland | Function |
|---|---|---|
| Insulin | Pancreas (islets of Langerhans) | Lowers blood glucose |
| Adrenaline | Adrenal gland | Prepares the body for "fight or flight" |
| Thyroxine | Thyroid | Controls metabolic rate; lack causes goitre (with iodine deficiency) |
| ADH | Pituitary | Increases water reabsorption in the kidney |
| FSH, LH | Pituitary | Control the menstrual cycle and ovulation |
| Auxin | Shoot and root tips (plants) | Growth; phototropism and geotropism |

## The nitrogen cycle

| Organism | Role | |
|---|---|---|
| Rhizobium | Nitrogen fixation in root nodules of legumes | $N_2 \rightarrow$ compounds |
| Nitrosomonas | Nitrification | Ammonium → nitrite |
| Nitrobacter | Nitrification | Nitrite → nitrate |
| Denitrifying bacteria | Denitrification | Nitrate → $N_2$ |
| Lightning | Fixes nitrogen into oxides, washed into the soil by rain | |

Genetics in one line: dominant alleles are written in capitals (T), recessive in small letters (t). A homozygous organism has two identical alleles (TT or tt); a heterozygous one has two different alleles (Tt). The genotype is the alleles; the phenotype is what you see.

## Part 3 Worked examples

**Example 1.** Two people with genotype AS have children. What fraction of the children are expected to be SS (sickle-cell disease)?

→ Gametes: A, S from each parent

→ Cross: AA, AS, AS, SS

→ Ratio 1 AA : 2 AS : 1 SS

**Answer:** 1/4 (25%) SS; 1/2 AS carriers; 1/4 AA

**Example 2.** A man of blood group A (genotype $I^AI^O$) marries a woman of group B ($I^BI^O$). What blood groups can their children have?

→ Gametes: $I^A$, $I^O$ and $I^B$, $I^O$

→ Children: $I^AI^B$ (AB), $I^AI^O$ (A), $I^BI^O$ (B), $I^OI^O$ (O)

**Answer:** AB, A, B or O, each with a chance of 1/4

**Example 3.** In peas, tall (T) is dominant to short (t). A heterozygous tall plant is crossed with a short plant. Give the ratio of the offspring.

→ Tt × tt; gametes T, t and t

→ Offspring: Tt, Tt, tt, tt

**Answer:** 1 tall : 1 short (a test cross)

**Example 4.** A woman who is a carrier of colour blindness ($X^cX$) marries a man with normal vision (XY). What are the chances for their children?

→ Gametes: $X^c$, X and X, Y

→ Children: $X^cX$ (carrier daughter), XX (normal daughter), $X^cY$ (colour-blind son), XY (normal son)

**Answer:** Half the sons colour-blind; half the daughters carriers; no colour-blind daughters

**Example 5.** Two tall plants (Tt × Tt) are crossed. What is the ratio of the phenotypes?

→ Offspring: TT, Tt, Tt, tt

→ 3 have T (tall), 1 is tt (short)

**Answer:** 3 tall : 1 short

**Example 6.** In a food chain, producers fix 10 000 kJ of energy. If about 10% passes to each level, how much reaches the secondary consumers?

→ Primary consumers: 10% of 10 000 = 1000 kJ

→ Secondary consumers: 10% of 1000 = 100 kJ

**Answer:** About 100 kJ

**Example 7.** A cell 0.05 mm long appears 20 mm long in a drawing. What is the magnification?

→ Magnification = image size ÷ actual size = 20 ÷ 0.05

**Answer:** × 400

**Example 8.** Which part of the nephron does ultrafiltration take place in?

→ Blood at high pressure in the glomerulus is filtered into Bowman's capsule

→ Selective reabsorption follows in the tubules

**Answer:** The glomerulus and Bowman's capsule

**Example 9.** Classify: a house fly, a spider, a millipede and a crab.

→ Insects: 6 legs, 3 body parts, often wings: house fly

→ Arachnids: 8 legs, 2 body parts: spider

→ Myriapods: many legs: millipede; Crustaceans: two pairs of antennae, hard shell: crab

**Answer:** All are arthropods, in four different classes

## Part 4 Traps that cost marks

| Trap | What to do |
|---|---|
| Confusing genotype and phenotype | Genotype is the alleles (Tt); phenotype is the visible trait (tall). Ratios of each can differ. |
| Thinking plants do not respire | Plants respire all the time, day and night; they photosynthesise only in light. |
| Arteries and veins | Arteries carry blood away from the heart (all oxygenated except the pulmonary artery); veins carry blood to the heart (all deoxygenated except the pulmonary vein). |
| Mixing up the enzymes | Learn enzyme, where it works, its substrate and its product as one set. |
| Pyramid of numbers vs energy | A pyramid of numbers can be inverted (one tree, many insects); a pyramid of energy never is. |
| Sex-linked traits in girls | A daughter shows a sex-linked recessive trait only if she inherits it from both parents. |
| Monocots and dicots | Monocots: one cotyledon, parallel veins, fibrous roots (maize). Dicots: two cotyledons, net veins, tap root (beans). |
| Malaria | Caused by Plasmodium (a protist), carried by the female Anopheles mosquito. The mosquito is the vector, not the cause. |
| Not reading "which is NOT" | Underline NOT, EXCEPT and LEAST before you answer. |

## Part 5 An 8-week plan

Start at least eight weeks before the exam. Each week: learn the topics, make short notes and solve past questions under time. Keep one day a week for revision.

| Week | Focus | Practice |
|---|---|---|
| 1 | Living things, the cell, levels of organisation, classification | Labelled diagrams; classification questions |
| 2 | Nutrition in plants and animals; enzymes; food tests | Digestion and food test questions |
| 3 | Transport in plants and animals; respiration | Heart, blood and transpiration questions |
| 4 | Excretion; support and movement; skeleton and joints | Kidney, skin and skeleton diagrams |
| 5 | Reproduction in plants and animals; growth | Flower, fruit and human reproduction questions |
| 6 | Coordination: nervous system, sense organs, hormones, plant responses | Eye, ear and hormone questions |
| 7 | Ecology; genetics; evolution | Genetic crosses and food chain calculations |
| 8 | Revision of weak areas; full timed mock exams | Two full CBT mocks; review every error |

### Exam-day reminders

Read every option before choosing. Eliminate the clearly wrong ones first.

Do not spend more than a minute on one question: flag it and come back.

Answer every question; there is no penalty for a wrong answer.

Check that your answers are saved before you submit, and keep an eye on the clock for all four subjects.

## Test Practice test

Give yourself 10 minutes for these 15 questions, as in the real exam. The answers follow.

**1. The organelle where aerobic respiration takes place is the:**

a) Ribosome  b) Mitochondrion  c) Nucleus  d) Chloroplast

**2. Which of these is found in plant cells but not in animal cells?**

a) Cell membrane  b) Nucleus  c) Cell wall  d) Mitochondrion

**3. The end product of protein digestion is:**

a) Amino acids  b) Glucose  c) Fatty acids  d) Glycerol

**4. A food that gives a violet colour with Biuret reagent contains:**

a) Starch  b) Fat  c) Reducing sugar  d) Protein

**5. The blood vessel that carries oxygenated blood from the lungs to the heart is the:**

a) Pulmonary artery  b) Aorta  c) Pulmonary vein  d) Vena cava

**6. Which hormone lowers blood sugar?**

a) Adrenaline  b) Insulin  c) Thyroxine  d) Glucagon

**7. Two carriers (AS × AS) have a child. The chance that the child is AS is:**

a) 1/4  b) 1/2  c) 3/4  d) 0

**8. A person with blood group O can receive blood only from group:**

a) A  b) B  c) AB  d) O

**9. Bacteria in the root nodules of legumes are involved in:**

a) Nitrogen fixation  b) Denitrification  c) Photosynthesis  d) Decomposition only

**10. In a food chain, the organisms with the most energy available are the:**

a) Producers  b) Primary consumers  c) Secondary consumers  d) Top carnivores

**11. Malaria is caused by:**

a) Anopheles  b) A virus  c) Plasmodium  d) Trypanosoma

**12. Which of these is an insect?**

a) Spider  b) Grasshopper  c) Millipede  d) Scorpion

**13. The theory of natural selection was proposed by:**

a) Lamarck  b) Mendel  c) Linnaeus  d) Darwin

**14. The functional unit of the kidney is the:**

a) Nephron  b) Neuron  c) Alveolus  d) Villus

**15. Water moves from a dilute solution to a concentrated one through a membrane by:**

a) Diffusion  b) Active transport  c) Osmosis  d) Transpiration

| No. | Answer | No. | Answer |
|---|---|---|---|
| 1 | b) Mitochondrion | 9 | a) Nitrogen fixation |
| 2 | c) Cell wall | 10 | a) Producers |
| 3 | a) Amino acids | 11 | c) Plasmodium |
| 4 | d) Protein | 12 | b) Grasshopper |
| 5 | c) Pulmonary vein | 13 | d) Darwin |
| 6 | b) Insulin | 14 | a) Nephron |
| 7 | b) 1/2 | 15 | c) Osmosis |
| 8 | d) O | | |

## Recommended textbooks

- **Modern Biology for Senior Secondary Schools**, S. T. Ramalingam.
- **Essential Biology**, M. C. Michael.
- Other texts listed in the current JAMB Biology syllabus.
- **Past questions**: solve at least the last ten years' UTME Biology questions, timing yourself, and study every one you get wrong.


### Prepare with the Himaayah Academy

Live UTME, Post-UTME and O-Level classes with a personal mentor, daily tasks, weekly mock exams, CBT practice and a leaderboard, all online.

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وَمَا تَوْفِيقِي إِلَّا بِاللَّهِ

"And my success is only through Allah." (Hūd 11:88)

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# Learn the living world, one system at a time.

This free guide from the Himaayah Academy helps UTME candidates prepare for Biology with a clear plan, the facts that come up every year and genetics explained step by step.

- The JAMB Biology syllabus in five areas
- Cells, nutrition, transport and coordination at a glance
- Genetics crosses, including the sickle-cell genotype
- Ecology, classification and food tests
- An 8-week plan and a 15-question practice test

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